Lowenheim-Skolem fails for second-order logic

The Open Logic Project contributors

openlogic:second-order-logic/metatheory:local-snapshot-20261013;
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Editorial note. The counted statement is the author theorem labelled thm:sol-no-ls (567-570); the proof span 572-584 is the author proof environment that immediately follows it.

Editorial scope. One original result of the Open Logic Project unit second-order-logic/metatheory, counted once: Loewenheim-Skolem fails for second-order logic. Only the counted theorem and its own author proof are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

Allowed standard prerequisites (editorial).

- The Count sentence. The second-order sentence Count, asserting existence of an inductive subset containing z and closed under u that covers the domain, is true in a structure exactly when its domain is enumerable; the unit establishes this before the counted theorem.. It converts the negated Count sentence into the assertion that the domain is nonenumerable (572-578).
- Nonenumerable sets exist. There are nonenumerable sets, for instance the power set of the naturals or the reals, as established by Cantor's diagonal argument.. It supplies the infinite models of the negated Count sentence (578-582).

Theorem 0.1. *The Löwenheim–Skolem Theorem fails for second-order logic: There are !!sentences with infinite models but no !!enumerable models.*

Proof. Recall that

$$\mathrm{fn}\, Count\, \mathrm{id}\, \exists[z][\exists[u][\forall[X][((X(z) \wedge \forall[x][(X(x) \to X(u(x)))]) \to \forall[x][X(x)])]]]$$

is true in !!astructure $\mathfrak{A}M$ iff $|\mathfrak{A}|M$ is !!enumerable, so $\neg\, \mathrm{fn}\, Count$ is true in $\mathfrak{A}M$ iff $|\mathfrak{A}|M$ is !!nonenumerable. There are such !!structures—take any !!nonenumerable set as the !!domain, e.g., $\mathcal{P}\mathbb{N}$ or $\mathbb{R}$. So $\neg\, \mathrm{fn}\, Count$ has infinite models but no !!enumerable models. □

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